Foundational Proofs of Measure Spaces

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1 Definitions

Definition 1. *Suppose X is a set and $\mathcal{S}$ is a set of subsets of X. Then $\mathcal{S}$ is called a σ-algebra on X if the following three conditions are satisfied:*
1. $\emptyset \in \mathcal{S}$
2. if $E \in \mathcal{S}$ *then* $X \backslash E \in \mathcal{S}$
3. if $E_1, E_2, \ldots$ *is a sequence of elements in* $\mathcal{S}$*, then* $\bigcup_{k=1}^{\infty} E_k \in \mathcal{S}$

Definition 2. *A measurable space is an ordered pair $(X, \mathcal{S})$ where X is a set and $\mathcal{S}$ is a σ-algebra on X. An element of $\mathcal{S}$ is called an $\mathcal{S}$-measurable set or just measurable if $\mathcal{S}$ is clear from the context.*

Definition 3. *Suppose X is a set and $\mathcal{S}$ is a σ-algebra on X. A measure on $(X, \mathcal{S})$ is a function $\mu \colon \mathcal{S} \to [0, \infty]$ such that $\mu(\emptyset) = 0$ and*

$$\mu\left(\bigcup_{k=1}^{\infty} E_k\right) = \sum_{k=1}^{\infty} \mu(E_k) \tag{1}$$

for every disjoint sequence $E_1, E_2, \ldots$ of $\mathcal{S}$.

Definition 4. *A measure space is an ordered triple $(X, \mathcal{S}, \mu)$, where X is a set, $\mathcal{S}$ is a σ-algebra on X, and μ is a measure on $(X, \mathcal{S})$.*

Remark 1. *One common measure space that mathematicians will deal with is the finite probability space $(\Omega, \mathscr{F}, P)$ where P is the probability measure. P has the added properties that $0 \leq P(E) \leq 1$ for every set $E \in \mathscr{F}$ and*

$$\sum_{\omega \in \Omega} P(\omega) = 1. \tag{2}$$

2 Theorems

Theorem 1. *Suppose $(X, \mathcal{S}, \mu)$ is a measure space $D, E \in \mathcal{S}$ such that $D \subseteq E$. Then:*
(a). $\mu(D) \leq \mu(E)$
(b). $\mu(E \backslash D) = \mu(E) - \mu(D)$ *provided* $\mu(D) < \infty$.

Proof. First we note that $E = D \cup (E \backslash D)$ and that D and $E \backslash D$ are disjoint. Thus $\mu(E) = \mu(D \cup (E \backslash D)) = \mu(D) + \mu(E \backslash D)$. Since by definition $\mu(E \backslash D) \geq 0$, it follows that $\mu(D) \leq \mu(E)$, proving part (a). For part (b) note that if $\mu(D) < \infty$ then $\mu(E) = \mu(D) + \mu(E \backslash D) \implies \mu(E \backslash D) = \mu(E) - \mu(D)$ as desired. $\square$

Theorem 2. *Suppose $(X, \mathcal{S}, \mu)$ is a measure space and $D, E \in \mathcal{S}$ with $\mu(D \cap E) < \infty$. Then*

$$\mu(D \cup E) = \mu(D) + \mu(E) - \mu(D \cap E). \tag{3}$$

Proof. First, note that $D \cup E = (D \backslash (D \cap E)) \cup E$ and that $D \backslash (D \cap E)$ and E are disjoint from each other. Thus $\mu(D \cup E) = \mu((D \backslash (D \cap E)) \cup E) = \mu(D \backslash (D \cap E)) + \mu(E) = \mu(D) - \mu(D \cap E) + \mu(E) = \mu(D) + \mu(E) - \mu(D \cap E)$ as desired. $\square$

Remark 2. *Theorem 2 when applied to the finite probability space $(\Omega, \mathscr{F}, P)$ gives us the addition theorem of probability $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. Since $0 \leq P(E) \leq 1$ for every set $E \in \mathscr{F}$, this subtraction is always valid.*

Theorem 3. *Suppose $(X, \mathcal{S}, \mu)$ is a measure space and $E_1, E_2, ... \in \mathcal{S}$. Then*

$$\mu(\bigcup_{k=1}^{\infty} E_k) \leq \sum_{k=1}^{\infty} \mu(E_k). \tag{4}$$

Proof. First we will show that

$$\bigcup_{k=1}^{\infty} E_k = \bigcup_{k=1}^{\infty} (E_k \backslash \bigcup_{i=1}^{k-1} E_i). \tag{5}$$

($\implies$) Suppose that x is in $\bigcup_{k=1}^{\infty} E_k$. Consider all sets E_i s.t. $x \in E_i$ and the set $I = \{i | x \in E_i\}$. Since $I \subseteq \mathbb{Z}^+$ and $\mathbb{Z}^+$ is well-ordered and countable, the set I has smallest element, which we will call j. Thus $x \in E_j \backslash \bigcup_{i=1}^{j-1} E_i$ since $x \notin E_i$ for $i < j$. Thus $x \in \bigcup_{k=1}^{\infty} (E_k \backslash \bigcup_{i=1}^{k-1} E_i)$.
($\impliedby$) Suppose $x \in \bigcup_{k=1}^{\infty} (E_k \backslash \bigcup_{i=1}^{k-1} E_i)$. Then $x \in E_k \backslash \bigcup_{i=1}^{k-1} E_i$ for some $k \in \mathbb{Z}^+$. Thus $x \in E_k \implies x \in \bigcup_{k=1}^{\infty} E_k$.
We now show that all sets $E_k \backslash \bigcup_{i=1}^{k-1} E_i$ are disjoint. Suppose for the sake of contradiction that $x \in E_k \backslash \bigcup_{i=1}^{k-1} E_i$ and $x \in E_j \backslash \bigcup_{i=1}^{j-1} E_i$ and WLOG suppose $k < j$. Then $x \in E_k$ since $x \in E_k \backslash \bigcup_{i=1}^{k-1} E_i$, but $x \notin E_k$ because $x \in E_j \backslash \bigcup_{i=1}^{j-1} E_i$. Therefore all sets $E_k \backslash \bigcup_{i=1}^{k-1} E_i$ are disjoint. Thus $\mu(\bigcup_{k=1}^{\infty} E_k) = \mu(\bigcup_{k=1}^{\infty} (E_k \backslash \bigcup_{i=1}^{k-1} E_i)) = \sum_{k=1}^{\infty} \mu(E_k \backslash \bigcup_{i=1}^{k-1} E_i) \leq \sum_{k=1}^{\infty} \mu(E_k)$. $\square$